Foundations of Denoising diffusion probabilistic model

[Denoising diffusion probabilistic model]

1.0 Content Block

To construct a probability path, we start by construct a conditional probability path p t (x | z) {\displaystyle p_{t}(x\vert z)} and the corresponding conditional velocity field v t (x | z) {\displaystyle v_{t}(x\vert z)} on some conditional distribution q (z) {\displaystyle q(z)} . A natural choice is the Gaussian conditional probability path:

2.0 Content Block

L s i m p l e , t = E x 0 , x t ~ q [■ ■ θ (x t , t) – x t – α ¯ t x 0 σ t ■ 2] = E x t ~ q , x 0 ~ q (· | x t) [■ ■ θ (x t , t) – x t – α ¯ t x 0 σ t ■ 2] {\displaystyle L_{simple,t}=E_{x_{0},x_{t}\sim q}\left[\left\|\epsilon _{\theta }(x_{t},t)-{\frac {x_{t}-{\sqrt {{\bar {\alpha }}_{t}}}x_{0}}{\sigma _{t}}}\right\|^{2}\right]=E_{x_{t}\sim q,x_{0}\sim q(\cdot |x_{t})}\left[\left\|\epsilon _{\theta }(x_{t},t)-{\frac {x_{t}-{\sqrt {{\bar {\alpha }}_{t}}}x_{0}}{\sigma _{t}}}\right\|^{2}\right]}

3.0 Content Block

Architectural improvements proposed various architectural improvements. For example, they proposed log-space interpolation during backward sampling. Instead of sampling from x t – 1 ~ N (μ ~ t (x t , x ~ 0) , σ ~ t 2 I) {\displaystyle x_{t-1}\sim {\mathcal {N}}({\tilde {\mu }}_{t}(x_{t},{\tilde {x}}_{0}),{\tilde {\sigma }}_{t}^{2}I)} , they recommended sampling from N (μ ~ t (x t , x ~ 0) , (σ t v σ ~ t 1 – v) 2 I) {\displaystyle {\mathcal {N}}({\tilde {\mu }}_{t}(x_{t},{\tilde {x}}_{0}),(\sigma _{t}^{v}{\tilde {\sigma }}_{t}^{1-v})^{2}I)} for a learned parameter v {\displaystyle v} . In the v-prediction formalism, the noising formula x t = α ¯ t x 0 + 1 – α ¯ t ■ t {\displaystyle x_{t}={\sqrt {{\bar {\alpha }}_{t}}}x_{0}+{\sqrt {1-{\bar {\alpha }}_{t}}}\epsilon _{t}} is reparameterised by an angle φ t {\displaystyle \phi _{t}} such that cos ■ φ t = α ¯ t {\displaystyle \cos \phi _{t}={\sqrt {{\bar {\alpha }}_{t}}}} and a "velocity" defined by cos ■ φ t ■ t – sin ■ φ t x 0 {\displaystyle \cos \phi _{t}\epsilon _{t}-\sin \phi _{t}x_{0}} . The network is trained to predict the velocity v ^ θ {\displaystyle {\hat {v}}_{\theta }} , and denoising is by x φ t – δ = cos ■ (δ) x φ t – sin ■ (δ) v ^ θ (x φ t) {\displaystyle x_{\phi _{t}-\delta }=\cos(\delta)\;x_{\phi _{t}}-\sin(\delta){\hat {v}}_{\theta }\;(x_{\phi _{t}})} . This parameterization was found to improve performance, as the model can be trained to reach total noise (i.e. φ t = 90 ■ {\displaystyle \phi _{t}=90^{\circ }}) and then reverse it, whereas the standard parameterization never reaches total noise since α ¯ t > 0 {\displaystyle {\sqrt {{\bar {\alpha }}_{t}}}>0} is always true.

4.0 Content Block

Diffusion model For generating images by DDPM, we need a neural network that takes a time t {\displaystyle t} and a noisy image x t {\displaystyle x_{t}} , and predicts a noise ■ θ (x t , t) {\displaystyle \epsilon _{\theta }(x_{t},t)} from it. Since predicting the noise is the same as predicting the denoised image, then subtracting it from x t {\displaystyle x_{t}} , denoising architectures tend to work well. For example, the U-Net, which was found to be good for denoising images, is often used for denoising diffusion models that generate images. For DDPM, the underlying architecture ("backbone") does not have to be a U-Net. It just has to predict the noise somehow. For example, the diffusion transformer (DiT) uses a Transformer to predict the mean and diagonal covariance of the noise, given the textual conditioning and the partially denoised image. It is the same as standard U-Net-based denoising diffusion model, with a Transformer replacing the U-Net. Mixture of experts-Transformer can also be applied. DDPM can be used to model general data distributions, not just natural-looking images. For example, Human Motion Diffusion models human motion trajectory by DDPM. Each human motion trajectory is a sequence of poses, represented by either joint rotations or positions. It uses a Transformer network to generate a less noisy trajectory out of a noisy one.

5.0 Content Block

N (μ , Σ) {\displaystyle {\mathcal {N}}(\mu ,\Sigma)} is the normal distribution with mean μ {\displaystyle \mu } and variance Σ {\displaystyle \Sigma } , and N (x | μ , Σ) {\displaystyle {\mathcal {N}}(x|\mu ,\Sigma)} is the probability density at x {\displaystyle x} . A vertical bar denotes conditioning. A forward diffusion process starts at some starting point x 0 ∼ q {\displaystyle x_{0}\sim q} , where q {\displaystyle q} is the probability distribution to be learned, then repeatedly adds noise to it by x t = 1 − β t x t − 1 + β t z t {\displaystyle x_{t}={\sqrt {1-\beta _{t}}}x_{t-1}+{\sqrt {\beta _{t}}}z_{t}} where z 1 , . . . , z T {\displaystyle z_{1},...,z_{T}} are IID (Independent and identically distributed random variables) samples from N (0 , I) {\displaystyle {\mathcal {N}}(0,I)} . The coefficients 1 − β t {\displaystyle {\sqrt {1-\beta _{t}}}} and β t {\displaystyle {\sqrt {\beta _{t}}}} ensure that Var (X t) = I {\displaystyle {\mbox{Var}}(X_{t})=I} assuming that Var (X 0) = I {\displaystyle {\mbox{Var}}(X_{0})=I} . The values of β t {\displaystyle \beta _{t}} are chosen such that for any starting distribution of x 0 {\displaystyle x_{0}} , if it has finite second moment, then lim t → ∞ x t | x 0 {\displaystyle \lim _{t\to \infty }x_{t}|x_{0}} converges to N (0 , I) {\displaystyle {\mathcal {N}}(0,I)} . The entire diffusion process then satisfies q (x 0 : T) = q (x 0) q (x 1 | x 0) ■ q (x T | x T − 1) = q (x 0) N (x 1 | α 1 x 0 , β 1 I) ■ N (x T | α T x T − 1 , β T I) {\displaystyle q(x_{0:T})=q(x_{0})q(x_{1}|x_{0})\cdots q(x_{T}|x_{T-1})=q(x_{0}){\mathcal {N}}(x_{1}|{\sqrt {\alpha _{1}}}x_{0},\beta _{1}I)\cdots {\mathcal {N}}(x_{T}|{\sqrt {\alpha _{T}}}x_{T-1},\beta _{T}I)} or ln ■ q (x 0 : T) = ln ■ q (x 0) − ∑ t = 1 T 1 2 β t ■ x t − 1 − β t x t − 1 ■ 2 + C {\displaystyle \ln q(x_{0:T})=\ln q(x_{0})-\sum _{t=1}^{T}{\frac {1}{2\beta _{t}}}\|x_{t}-{\sqrt {1-\beta _{t}}}x_{t-1}\|^{2}+C} where C {\displaystyle C} is a normalization constant and often omitted. In particular, we note that x 1 : T | x 0 {\displaystyle x_{1:T}|x_{0}} is a Gaussian process, which affords us considerable freedom in reparameterization. For example, by standard manipulation with Gaussian process, x t | x 0 ∼ N (α ¯ t x 0 , σ t 2 I) {\displaystyle x_{t}|x_{0}\sim N\left({\sqrt {{\bar {\alpha }}_{t}}}x_{0},\sigma _{t}^{2}I\right)}